

THE TALE OF TWO BREASTS: SYNCHRONOUS RIGHT BREAST LYMPHOMA AND LEFT BREAST INVASIVE CARCINOMA

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BACKGROUND

Synchronous bilateral breast malignancies of different histological subtypes are rare and pose diagnostic and therapeutic challenges. Here, we report a case of advanced left breast invasive carcinoma and right breast diffuse large B-cell lymphoma managed with multidisciplinary team.

CASE PRESENTATION

A 55-year-old woman presented with an ulcerative left breast mass and a right breast lump.

Mammography and ultrasound demonstrated bilateral suspicious breast masses with bilateral axillary lymphadenopathy (BI-RADS 5). Computed tomography for staging showed advanced disease with pulmonary and bone metastases.

Histopathology revealed two distinct malignancies: Left breast invasive carcinoma and right breast diffuse large B-cell lymphoma (DLBCL). Right axillary lymph node biopsy yielded metastatic carcinoma, likely from the contralateral breast.

The patient was referred to haematology team for further management of the lymphoma.

DISCUSSION

Breast lymphoma accounts for less than 1% of breast malignancies, and its synchronous occurrence with invasive breast carcinoma is exceptionally uncommon. Breast lymphoma may clinically resemble breast carcinoma, commonly presenting as a breast lump or axillary lymphadenopathy. The imaging findings (mammogram and ultrasound) of breast lymphoma are nonspecific and mimic those of breast carcinoma. While invasive breast carcinoma is commonly managed with a combination of surgery and receptor-directed systemic therapy, breast lymphoma is primarily treated with systemic chemo-immunotherapy and radiotherapy. Managing synchronous malignancies is challenging and requires multidisciplinary prioritisation to determine the optimal treatment sequence.

FIGURES

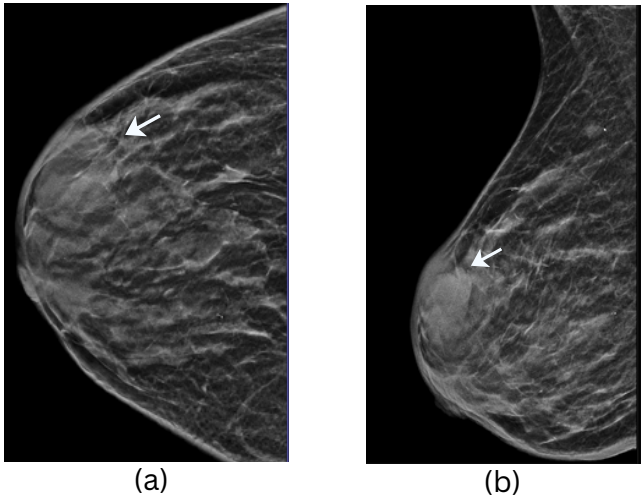


Figure 1: Right mammogram (a) craniocaudal and (b) mediolateral oblique views show lobulated, circumscribed, equal density mass in the right upper outer quadrant (arrows).

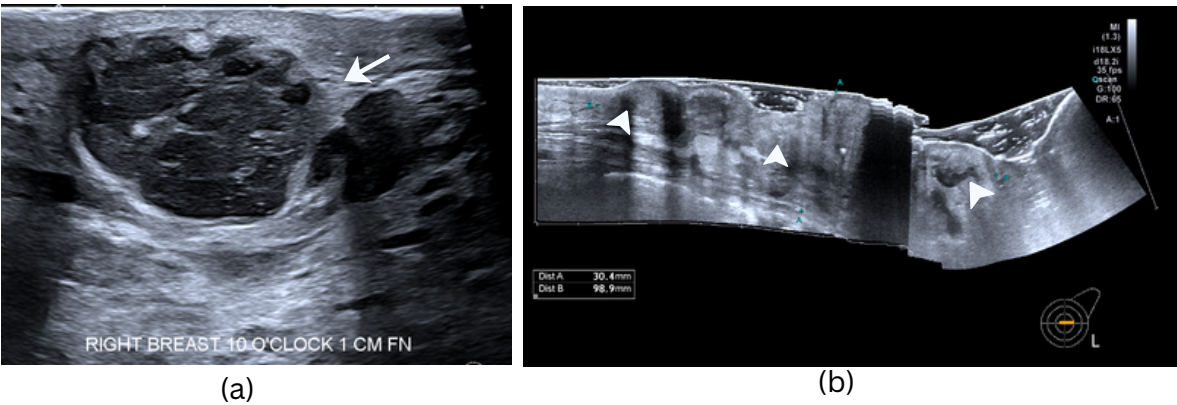


Figure 2: Ultrasound findings (a) lobulated, circumscribed, hypoechoic mass at right 10:00 1cmFN (arrow) (b) irregular, indistinct, hypoechoic mass occupying the left breast with extension to the overlying skin (arrowheads).

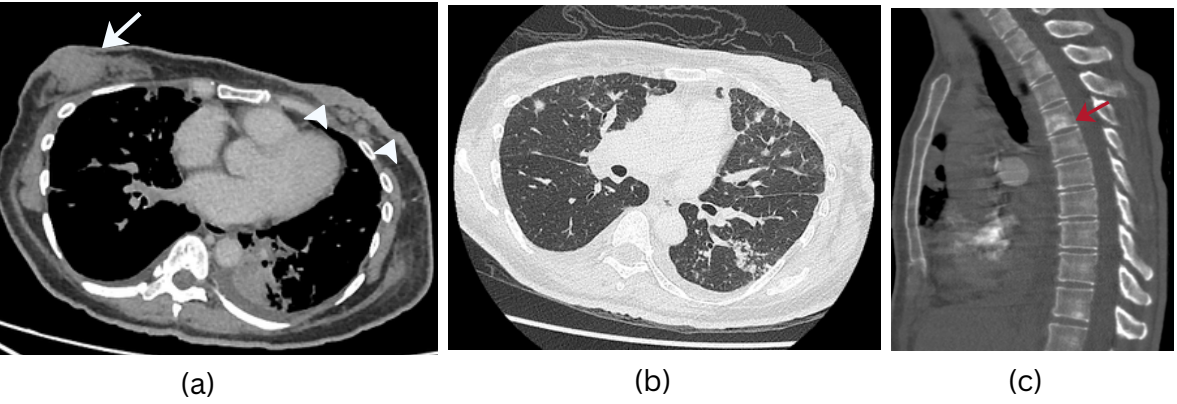


Figure 3: CECT findings (a) lobulated, heterogeneously enhancing lesion in the right breast (arrow) and irregular, heterogeneously enhancing mass infiltrating the skin in the left breast (arrowheads) (b) multiple lung nodules in keeping with pulmonary metastasis (c) bone metastasis (red arrow).

CONCLUSION / TAKE-HOME MESSAGE

Synchronous breast lymphoma and invasive breast carcinoma, although rare, should be considered in patients presenting with bilateral suspicious breast lesions. Multimodality imaging and histopathological confirmation of each suspicious lesion are crucial for avoiding diagnostic pitfalls and ensuring appropriate multidisciplinary management.

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